Worksheet 4.1 — Logarithms & the Laws of Logarithms

Grade 12 Advanced Functions (MHF4U) · Unit 4: Exponential and Logarithmic Functions

The Big Ideas

A logarithm answers "what exponent?": $\log_b x = y \iff b^y = x$. The laws turn products, quotients and powers into sums, differences and multiples. In every step write the **whole expression or equation**, and name the law you use.

- Switch forms: $\log_b x = y \iff b^y = x$; the base stays the base and the exponent is the logarithm. The argument x must be positive.
- Product $\log_b(xy) = \log_b x + \log_b y$; Quotient $\log_b \frac{x}{y} = \log_b x - \log_b y$; Power $\log_b(x^n) = n \log_b x$.
- Change of base: $\log_b x = \dfrac{\log x}{\log b}$. Mistakes to avoid: $\log(x+y) \neq \log x + \log y$.

Core Concepts

A logarithm is an exponent

$\log_b x$ is the power to which b must be raised to give x the inverse operation of exponentiation.

The laws

Logs turn products, quotients, and powers into sums, differences, and multiples: $\log_b(MN) = \log_b M + \log_b N$ and $\log_b M^p = p \log_b M$.

Change of base

Evaluate any logarithm on a calculator with $\log_b x = \dfrac{\log x}{\log b}$.

Writing solutions

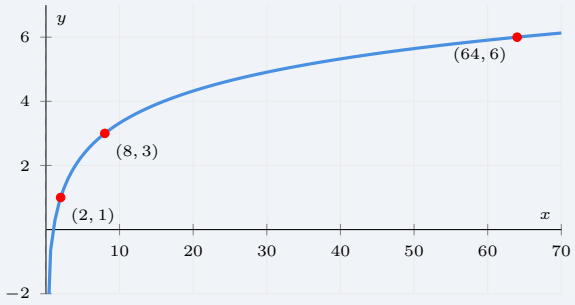
In every step write the **whole expression**, for example $\log_2(8 \cdot 16) = \log_2 8 + \log_2 16$, and name the law you use (product, quotient, power, change of base). The argument of a logarithm must always be positive.

Worked Examples

Example 1 — Evaluate

Find $\log_2 64$.

Solution. **Step 1 (ask the question):** $\log_2 64$ asks: 2 to what power gives 64?
Step 2 (exponential form): Write $\log_2 64 = y$ as $2^y = 64$.
Step 3 (solve): Since $64 = 2^6$, the equation $2^y = 64$ becomes $2^y = 2^6$, so $y = 6$.
Step 4 (check): $2^6 = 64$ ✓; the point $(64, 6)$ is on the graph of $y = \log_2 x$.
Answer: $\log_2 64 = 6$.



Example 2 — Convert to log form

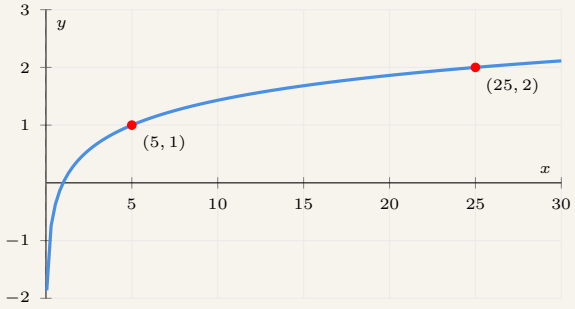
Write $10^4 = 10000$ in logarithmic form.

Solution. **Step 1 (identify the parts):** In $10^4 = 10000$ the base is 10, the exponent is 4 and the result is 10000.
Step 2 (logarithmic form): The exponent is the logarithm and the result is the argument: $\log_{10} 10000 = 4$, i.e. $\log 10000 = 4$ (no written base means base 10).
Step 3 (check): Reverse it: $\log_{10} 10000 = 4 \iff 10^4 = 10000$ ✓.
Answer: $\log 10000 = 4$.

Example 3 — Evaluate

Find $\log_5 25$.

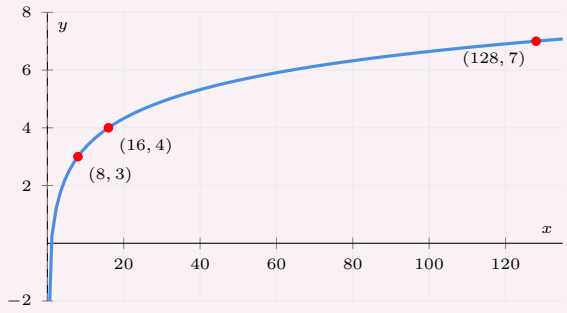
Solution. **Step 1 (exponential form):** Write $\log_5 25 = y$ as $5^y = 25$.
Step 2 (solve): Since $25 = 5^2$, $5^y = 25$ becomes $5^y = 5^2$, so $y = 2$.
Step 3 (check): $5^2 = 25$ ✓; the points $(5, 1)$ and $(25, 2)$ are on $y = \log_5 x$.
Answer: $\log_5 25 = 2$.



Example 4 — Product law

Simplify $\log_2(8 \cdot 16)$.

Solution. **Step 1 (product law):** $\log_2(8 \cdot 16) = \log_2 8 + \log_2 16$.
Step 2 (evaluate each term): $\log_2 8 + \log_2 16 = 3 + 4 = 7$, because $2^3 = 8$ and $2^4 = 16$.
Step 3 (check): $8 \cdot 16 = 128 = 2^7$, so $\log_2 128 = 7$ ✓.
Answer: $\log_2(8 \cdot 16) = 7$.



Example 5 — Power law

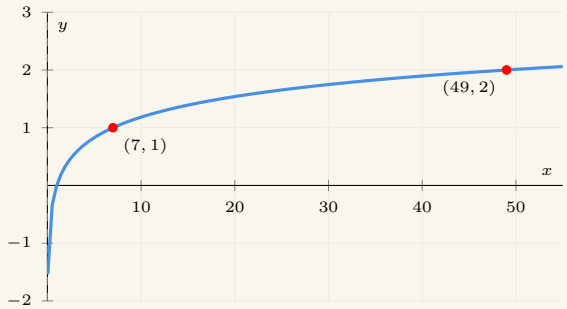
Simplify $\log_3(9^3)$.

Solution. **Step 1 (power law):** $\log_3(9^3) = 3\log_3 9$.
Step 2 (evaluate): $3\log_3 9 = 3 \cdot 2 = 6$, because $3^2 = 9$.
Step 3 (check): $9^3 = (3^2)^3 = 3^6$, so $\log_3(3^6) = 6$ ✓.
Answer: $\log_3(9^3) = 6$.

Example 6 — Evaluate

Find $\log_7 49$.

Solution. **Step 1 (exponential form):** Write $\log_7 49 = y$ as $7^y = 49$.
Step 2 (solve): Since $49 = 7^2$, $7^y = 7^2$ gives $y = 2$.
Step 3 (check): $7^2 = 49$ ✓; the point $(49, 2)$ is on $y = \log_7 x$.
Answer: $\log_7 49 = 2$.



Example 7 — Base 10

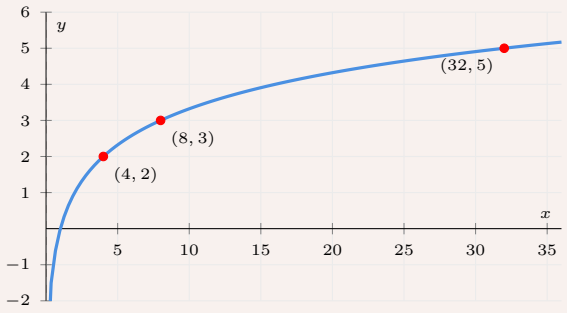
Find $\log 100000$.

Solution. **Step 1 (base):** $\log 100000$ has no written base, so it is $\log_{10} 100000$.
Step 2 (exponential form): Write $\log_{10} 100000 = y$ as $10^y = 100000 = 10^5$, so $y = 5$.
Step 3 (check): Count the zeros: 100000 has 5 zeros, so $\log 100000 = 5$ ✓.
Answer: $\log 100000 = 5$.

Example 8 — Quotient law

Simplify $\log_2 \dfrac{32}{4}$.

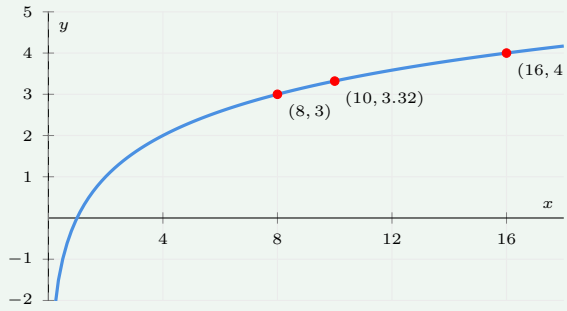
Solution. **Step 1 (quotient law):** $\log_2 \dfrac{32}{4} = \log_2 32 - \log_2 4$.
Step 2 (evaluate): $\log_2 32 - \log_2 4 = 5 - 2 = 3$.
Step 3 (check): $\dfrac{32}{4} = 8 = 2^3$, so $\log_2 8 = 3$ ✓.
Answer: $\log_2 \dfrac{32}{4} = 3$.



Example 9 — Change of base

Express $\log_2 10$ with base-10 logs and evaluate it.

Solution. **Step 1 (change of base):** $\log_2 10 = \dfrac{\log 10}{\log 2}$.
Step 2 (calculate): $\dfrac{\log 10}{\log 2} = \dfrac{1}{0.3010} \approx 3.32$.
Step 3 (check): $2^3 = 8 < 10 < 16 = 2^4$, so an exponent between 3 and 4 is reasonable ✓.
Answer: $\log_2 10 \approx 3.32$.



Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Evaluate $\log_2 16$.

Question 2

Write $3^4 = 81$ in log form.

Question 3

Evaluate $\log 10^6$.

Question 4

Simplify $\log_2(16 \cdot 32)$.

Question 5

Simplify $\log_2(4^5)$.

Question 6

Evaluate $\log_4 64$.

Question 7

Simplify $\log_2 \frac{64}{8}$.

Question 8

Evaluate $\log_2 1$.

Question 9

Express $\log_3 20$ with base-10 logs.

Question 10

Simplify $\log_b b^7$.

Question 11

Evaluate $\log_{10} 0.01$.

Question 12

Simplify $\log 2 + \log 5$.

Question 13 — Challenge

Write $\log_2 40$ as a sum/difference of $\log_2 2, \log_2 5$ (note $40 = 2^3 \cdot 5$).

Question Recap & Answer Key

Question Recap

1. Evaluate $\log_2 16$.
2. Write $3^4 = 81$ in log form.
3. Evaluate $\log 10^6$.
4. Simplify $\log_2(16 \cdot 32)$.
5. Simplify $\log_2(4^5)$.
6. Evaluate $\log_4 64$.
7. Simplify $\log_2 \dfrac{64}{8}$.
8. Evaluate $\log_2 1$.
9. Express $\log_3 20$ with base-10 logs.
10. Simplify $\log_b b^7$.
11. Evaluate $\log_{10} 0.01$.
12. Simplify $\log 2 + \log 5$.
13. Write $\log_2 40$ as a sum/difference of $\log_2 2, \log_2 5$ (note $40 = 2^3 \cdot 5$).

✓ Answer Key

1. $\log_2 16 = y \iff 2^y = 16 = 2^4$, so $y = 4$
2. base 3, exponent 4, result 81: $\log_3 81 = 4$
3. $\log 10^6 = \log_{10} 10^6 = 6$, since $10^6 = 10^6$
4. $\log_2(16 \cdot 32) = \log_2 16 + \log_2 32 = 4 + 5 = 9$
5. $\log_2(4^5) = 5 \log_2 4 = 5 \cdot 2 = 10$
6. $\log_4 64 = y \iff 4^y = 64 = 4^3$, so $y = 3$
7. $\log_2 \dfrac{64}{8} = \log_2 64 - \log_2 8 = 6 - 3 = 3$
8. $\log_2 1 = y \iff 2^y = 1 = 2^0$, so $y = 0$
9. $\log_3 20 = \dfrac{\log 20}{\log 3}$ (base-10 logs)
10. $\log_b b^7 = 7 \log_b b = 7 \cdot 1 = 7$
11. $\log_{10} 0.01 = y \iff 10^y = 0.01 = 10^{-2}$, so $y = -2$
12. $\log 2 + \log 5 = \log(2 \cdot 5) = \log 10 = 1$
13. $\log_2 40 = \log_2(2^3 \cdot 5) = \log_2 2^3 + \log_2 5 = 3 \log_2 2 + \log_2 5 = 3 + \log_2 5$